

Pouyan Navard

📍 Columbus, OH ✉ bnd.pouyan@gmail.com 🔗 bnavard.github.io 📄 Google Scholar

Skills

Generative AI & Diffusion Models: Video Diffusion Models, Diffusion Transformers (DiT), Stable Diffusion, ControlNet, Synthetic Data Generation, LoRA/QLoRA/PEFT, DPO for Generative Models, Image/Video/3D Generation, 3D Morphable Models (FLAME)

Vision & Multimodal AI: Vision Transformers, Vision-Language Models (LLaVA), Masked Multimodal Modeling, 3D Perception, Point Cloud Processing, Physical AI

Frameworks & Infrastructure: PyTorch, HuggingFace, PyTorch Lightning, CUDA, DDP/FSDP/DeepSpeed, NCCL, Multi-Node GPU Training, Hydra, OpenCV, Open3D, SLURM, AWS, Weights & Biases, Git

Languages: Python, C++, Bash

Industry Experience

Machine Learning Engineer

Turing

Remote (Columbus, OH)

Jun 2026 – Present

- Designed a **LangChain-based deep research agent** that automates benchmark quality review by grounding tasks in scientific literature—evaluating prompt ambiguity, instruction–test misalignment, and test coverage against authoritative sources to catch ungrounded, auto-generated tasks.
- Reviewed **200+ agentic tasks** across frontier-model benchmarks spanning scientific domains, machine learning, and software engineering—auditing prompt ambiguity, instruction–test misalignment, and test coverage through agent trajectory analysis, loss pattern analysis, and task inspection.
- Contributing to the design of a new **long-horizon SWE benchmark** for evaluating agentic AI on extended, multi-step software engineering tasks.

Computer Vision Engineer

Path Robotics Inc

Columbus, Ohio

Nov 2024 – May 2026

- Architected and trained an ~800M-parameter **multimodal foundation model** via cross-modal masked modeling on paired point cloud and image data across multi-node GPU clusters (DDP/FSDP), consolidating 5 per-customer models into a single generalizable model with **96%** classification accuracy.
- Post-trained the foundation model via **LoRA/QLoRA parameter-efficient fine-tuning** for domain adaptation to new customers, adapting task-specific heads to per-application requirements without full retraining.
- Quantized the model to INT8** for on-device inference on **NVIDIA Jetson**, cutting the memory footprint ~4× and enabling real-time edge deployment on robotic welding systems without GPU-server dependency.

Selected Publications

[Loki: Representation over Architecture for Diffusion-Based Portrait Animation](#) [🔗](#)

Pouyan Navard, Sernam Lim

Conference on Neural Information Processing Systems (NeurIPS), 2026 (submitted)

- Developed a **video diffusion model** for portrait animation that conditions on FLAME parametric face representations instead of RGB, achieving controllable cross-identity reenactment without cross-ID training data while requiring **43% fewer** parameters and **1,496× less** training video than the leading baseline. Trained on 8×H200 GPUs.
- Introduced two evaluation metrics (Head Pose Follow, Expression Follow) grounded in the parametric face model, exposing systematic gaps in the field’s inherited metric suite.

[KnobGen: Controlling the Sophistication of Artwork in Sketch-Based Diffusion Models](#) [🔗](#)

Pouyan Navard, Amin Monsefi, Mengxi Zhou, Wei-Lun Chao, Alper Yilmaz, Rajiv Ramnath

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Workshop, 2025

- Proposed a dual-pathway diffusion framework for **multi-condition controllable** image generation from

sketches, with coarse-grained and fine-grained spatial controllers linked by an adjustable knob mechanism, enabling high-quality synthesis from both novice and expert inputs.

SeamCam: Quantifying Seamless Camouflage via Multi-Cue Visual Detectability

Amin Monsefi et al.

European Conference on Computer Vision (ECCV), 2026 (under review)

- Applied **Direct Preference Optimization (DPO) fine-tuning** to diffusion-based inpainting models for camouflage image generation; introduced a camouflage evaluation metric achieving 78.82% agreement with human judgments and a 1.5k-image benchmark.

Large Language-and-Vision Assistant for Lunar Exploration (LLaVA-LE) [↗](#)

Gokce Inal, Pouyan Navard, Alper Yilmaz

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Workshop, 2026

- Fine-tuned LLaVA for lunar surface characterization via two-stage instruction tuning, achieving a **3.3×** performance gain; curated LUCID, a 96k-image multimodal dataset with 81k QA pairs from NASA archives.

SegFormer3D: An Efficient Transformer for 3D Medical Image Segmentation [↗](#)

Shehan Perera, Pouyan Navard, Alper Yilmaz

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Workshop, 2024

- Designed a lightweight hierarchical transformer achieving competitive accuracy with **33×** **fewer** parameters and **13×** **lower** compute than state-of-the-art 3D segmentation models.

Full publication list: [Google Scholar](#) [↗](#)

Awards and Honors

Robert E. Altenhofen Memorial Scholarship Award [↗](#)

ISPRS, 2022

- National-level scholarship recognizing excellence in photogrammetry and remote sensing research.

Education

The Ohio State University

Feb 2021 – Aug 2025

PhD in Electrical and Computer Engineering

- GPA: 3.9/4.0
- Dissertation: *Distribution-Aware Learning for Sparsely Labeled and Imbalanced Spatiotemporal Data*
- Research focus on 3D perception, diffusion-based generative modeling, and multimodal learning
- Supervisor: Prof. Alper Yilmaz.

Isfahan University

Sep 2015 – Sep 2017

MSc in Electrical and Computer Engineering

- GPA: 3.9/4.0
- Thesis: Geometric Structure from Motion under Large View Point Change using Epipolar Geometry

Isfahan University

Sep 2011 – Sep 2015

BSc in Electrical and Computer Engineering

- GPA: 3.6/4.0
- Thesis: 3D View Photogrammetric Reconstruction